Skillsheet D | ECON 0100 | Fall 2026

Strategic Interaction

How to use this sheet

This sheet lists every assessed skill in Part D. Your grade is the percentage of skills you pass across the semester, so this sheet is the container for your studying: it tells you what each skill is, where we build it, what practice unlocks it, and the standard you must meet to pass it on Checkpoint D.

Each skill has three types of practice: the **Exercise** (done together in class), the **Vignette** (done together in recitation), and the **Homework** (done on your own time, due Sundays). Practice is graded for completion, not correctness. But you will get feedback on which Homework questions you got right and wrong.

- Complete **2 of 3** practices for a skill to unlock the skill on Checkpoint D.
- Complete **3 of 3** practices for a skill to unlock the Reattempt.

Skills you pass stay passed. If you do not pass a skill on Checkpoint D, complete all three practices and take the Reattempt.

Assessment scope. The standards below define the evidence required for each pass. Show the payoff comparisons behind each answer; in a payoff matrix the first number in each cell belongs to the row player. Skills are judged separately. Games have two players with two strategies each. Public-goods questions give each person's benefit in a table.

The skills at a glance

Code	Skill	Practice
D1.1	Best Responses	Exercise D1 · Vignette D1 · HW D1
D1.2	Nash Equilibrium & Efficiency	Exercise D1 · Vignette D1 · HW D1
D2.1	Common Resources	Exercise D2 · Vignette D2 · HW D2

Code	Skill	Practice
D3.1	Public Goods	Exercise D3 · Vignette D3 · HW D3
D3.2	Voting on Public Goods	Exercise D3 · Vignette D3 · HW D3

D1.1 | Best Responses

When your payoff depends on what someone else does, your best choice can depend on theirs.

Standard. You pass this skill if you can:

- Hold the other player's choice fixed and compare your own payoffs to find your best response to each of their choices, including a tie where both choices are best responses.
- Identify a dominant strategy, a choice that is a best response to everything the other player could do, or explain why a player has none.

D1.2 | Nash Equilibrium & Efficiency

A Nash equilibrium is a pair of choices that are best responses to each other, and it need not be best for the group.

Standard. You pass this skill if you can:

- Find every Nash equilibrium in a payoff matrix, recognizing that a game can have one or more than one.
- Find the socially efficient outcome or outcomes by comparing total payoffs.
- Explain why the Nash equilibrium is or is not efficient, as in a Prisoner's Dilemma where each player's own interest leads both to an outcome worse than cooperating.

D2.1 | Common Resources

A common resource is rival but non-excludable, so private incentives lead to overuse.

Standard. You pass this skill if you can:

- Identify a common resource as rival and non-excludable.
- In a game over a shared resource, find each user's best responses, the Nash equilibrium, and the socially efficient outcome, and explain why the equilibrium overuses the resource: a user who takes more does not count the loss to the other user.
- Find the new Nash equilibrium when an enforced rule, such as a permit with a fine, changes the payoffs, and explain how the rule makes sustainable use a best response.

D3.1 | Public Goods

A public good is non-rival and non-excludable, so people can enjoy it without paying for it.

Standard. You pass this skill if you can:

- Identify a public good as non-rival and non-excludable, and explain why self-interested people will not contribute to it voluntarily (free riding).
- Calculate marginal social benefit by adding each person's marginal benefit, and compare it with the cost to decide whether providing the good is efficient.
- Show why a seller charging one price or fee to everyone cannot cover the cost, even when providing the good is efficient.

D3.2 | Voting on Public Goods

A vote can fund a public good, but a majority need not choose the efficient outcome.

Standard. You pass this skill if you can:

- Compare each person's benefit with a proposed tax or fee to determine how they vote, and whether the proposal wins a majority.
- Compare the vote's result with the efficient choice (marginal social benefit against cost), and explain how a vote can fail to fund an efficient good or impose a cost some voters do not value.